

Comparative Analysis of Models for Flight Delay Tracking and Prediction

William Bartels and Karen Works
Department of Computer Science
Florida State University
Panama City, FL 32405
`wrb14@fsu.edu`, `keworks@fsu.edu`

1 Introduction

In recent years, airline travel in the United States has become unreliable [2]. In an effort to give travelers more control and information about flight delays, we modeled flight delays in a two-phase approach. In the first phase, we looked at designing an efficient model to plot the effect a single flight delay has on later scheduled flights. The second phase looked at machine learning models to predict flight delays before they occur.

2 Dataset

The study used a comprehensive Kaggle flight delay data set that contains 484,551 domestic US flights from 2019 covering 12 airlines across 274 airports with 3,377 unique routes [1]. The data set includes detailed flight information such as scheduled and actual departure/arrival times, carrier details, aircraft tail numbers, origin and destination airports, and specific delay categorizations.

3 Results and Discussion

The first phase is a comparative performance analysis between graphical (Neo4j) and traditional relational (SQLite) databases to store and query the effects of a single flight delay across the scheduled upcoming flights measured query execution time. Our results show significant differences in query execution times between SQLite and Neo4j as dataset size scales. A query to track each plane across an entire day was ran times using 3 different datasets that varied by size, namely, datasets containing 10, 100, and 1000 distinct planes. For small

datasets (10 tail numbers, 665 flights), SQLite was superior with an average execution time of 0.45 seconds compared to Neo4j’s 2.05 seconds. However, as the dataset size increased Neo4j’s execution times surpassed SQLite. In the large dataset (1000 tail numbers, 127,239 flights) Neo4j’s execution times averaged 3.47 seconds while SQLite’s execution time dramatically increased to 40.87 seconds, making Neo4j $11.8\times$ faster. These results demonstrate that graph databases like Neo4j provide superior scalability and performance for queries involving self referencing relationships.

The second phase evaluates three machine learning models for delay prediction. Random Forest, Gradient Boosting, and XGBoost. These models were trained on flight data that include basic flight information such as departure times, carrier details, and route distances. The study performed feature engineering to create additional predictors, such as weekend flight indicators and time-based features. Performance comparison between the three models was made using metrics such as RMSE and prediction accuracy within various time thresholds. Feature importance analysis in all three models helped identify the most crucial factors in predicting flight delays.

To enhance the predictive capability of the machine learning models, comprehensive feature engineering was performed to extract meaningful temporal and operational patterns from the raw flight data. The departure time was decomposed into separate hour and minute components to capture time-of-day effects on delays, while a weekend indicator variable was created to account for different operational patterns between weekdays and weekend flights.. This feature engineering approach was designed to provide the models with more granular information about flight characteristics that are known to influence delay propagation in aviation operations.

The comparative analysis revealed that XGBoost achieved the best performance with an RMSE of 51.23 minutes, MAE of 32.33 minutes, and R^2 of 0.21, followed by Gradient Boosting and Random Forest. While short-term precision remained challenging (9.2% accuracy within ± 5 minutes), the models showed practical utility for operational planning with 62.0% accuracy within ± 30 minutes. Feature importance analysis identified departure hour, flight distance, and carrier as key predictors, with the models accurately capturing average delay patterns despite the inherent variability in flight operations.

4 Conclusion

This research provides a comprehensive framework for flight delay management by identifying an optimal database design for tracking delays and developing accurate prediction models for future delays.

References

- [1] Kaggle: Flight Delay and Causes.
- [2] Allison Morrow. Analysis: Travel in america is a crapshoot, and it won't get better anytime soon | cnn business, Jun 2023.